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Neural Network Course Project

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AlexNet: The Deep Learning Revolution

1. Aim of the Work

This project aims to demonstrate a clear, deep, and accessible understanding of AlexNet one of the most pivotal architectures in the history of deep learning. The primary objective is not merely to describe the model, but to communicate why it mattered: what problems it solved, what innovations it introduced, and what legacy it left behind for the field of computer vision and artificial intelligence as a whole.

The project was designed to fulfil three educational goals. First, to consolidate the team's own understanding of Convolutional Neural Networks by researching AlexNet from multiple angles architectural, historical, and mathematical. Second, to develop the skill of using AI tools effectively as assistants rather than substitutes, learning how to prompt, evaluate, and refine AI-generated content. Third, to practise science communication by translating a complex technical concept into a format that is visually engaging and easy to grasp within a few minutes.

The concept was chosen for its historical significance: AlexNet's victory at the ILSVRC 2012 competition is widely regarded as the starting point of the modern deep learning era, making it an ideal subject for demonstrating both the technical depth and the broader context of ANN development.

2. Deliverables Overview

The project produced four main deliverables as required:

#	Deliverable	Description
1	**Presentation (PPTX)**	A 24-slide visual presentation covering AlexNet's history, architecture, innovations (ReLU, Dropout, Overlapping Pooling, LRN, Data Augmentation, Multi-GPU), performance results, limitations, and comparison with LeNet and VGGNet.
2	**Simulation / Demo**	An interactive explainer video (7-8 min) generated with NotebookLM AI, covering AlexNet's architecture and key innovations in a conversational podcast format.
3	**MCQ Set (20 Questions)**	Twenty multiple-choice questions covering all major aspects of AlexNet architecture, innovations, historical context, training techniques each with four options, a clearly marked correct answer, and an explanation.
4	**Project Report (this document)**	A structured report explaining the aim, methodology, challenges, AI tool usage, and final conclusions of the project.

3. Methodology : Steps Followed

The team followed a structured five-phase workflow to produce all project components. The block diagram below summarises the steps:

Phase	Step	Description
Phase 1	**Concept Selection & Research**	Selected AlexNet as the topic. Conducted research using primary sources (the original Krizhevsky et al. 2012 paper), secondary sources (textbooks, online lectures), and AI tools to build a thorough understanding of the architecture and its context.
Phase 2	**Content Design & Structuring**	Planned the narrative arc of the presentation: from historical context → architecture → innovations → results → limitations → legacy. Divided the content into logical slides with clear flow. Drafted the MCQ questions aligned to learning objectives.
Phase 3	**Presentation Development**	Built the 24-slide PPTX using a consistent visual design. Used diagrams, tables, and structured layouts to make technical content visually accessible. Each innovation was given its own dedicated slide(s) to avoid overcrowding.
Phase 4	**Simulation / Video Production**	Used NotebookLM to generate an AI-hosted explanatory podcast script. Provided a detailed prompt specifying the concept, narrative order, analogies required, and the target audience. Reviewed and validated all AI-generated content for accuracy before finalising.
Phase 5	**MCQ & Report Creation**	Generated 20 MCQs covering all key aspects of AlexNet. Wrote this report to document methodology, challenges, and conclusions. Validated all AI-assisted outputs against primary sources.

4. Project Components

4.1. Presentation

The presentation consists of 24 slides organised into six thematic sections: Introduction, Historical Context, Architecture Deep-Dive, Key Innovations, Results & Legacy, and Limitations/Comparison. The design prioritises clarity by dedicating individual slides to each major innovation rather than listing them together, and by using diagrams, tables, and large typographic callouts (such as the '60M parameters' stat slide) to make numbers meaningful at a glance.

Key content highlights:

- Slide 4 presents the 8-layer architecture with a breakdown of convolutional and fully connected layers.
- Slides 5–6 explain ReLU activation, including the quantitative speed advantage (6× faster than Tanh).
- Slide 7 covers the dual-GPU training setup and its necessity given 2012 hardware constraints.

- Slides 8–11 address Overlapping Pooling, LRN, Data Augmentation, and Dropout respectively.
- Slides 12–14 drill into the per-layer architecture (Conv layers 1–5, FC layers 6–8).
- Slide 17 presents the ILSVRC 2012 results and the 10.8% margin of victory.
- Slides 18–20 compare AlexNet with LeNet (predecessor) and VGGNet (successor) and discuss limitations.

4.2. Simulation / Interactive Demo

The simulation deliverable is a 7–8 minute AI-generated explainer video produced using NotebookLM. The tool was prompted with a detailed specification that directed the AI hosts to cover: (1) the problem AlexNet solved, (2) the architecture walkthrough, (3) the three key innovations ReLU, Dropout, and Data Augmentation (4) the dual-GPU training context, and (5) AlexNet's role as a turning point in computer vision.

The conversational podcast format was selected because it naturally mirrors how an intuitive explanation would unfold one concept at a time, with transitions that relate each idea to the previous one. This format is well-suited for a 7–8 minute window because it avoids the abrupt cuts of a slideshow while maintaining momentum.

4.3. MCQ Set

Twenty multiple-choice questions were designed to cover the full breadth of AlexNet content. The questions span six cognitive categories: factual recall (e.g., publication year, number of layers), conceptual understanding (e.g., what ReLU solves, what dropout does), applied reasoning (e.g., why two GPUs were used), comparative knowledge (e.g., AlexNet vs LeNet), quantitative facts (e.g., top-5 error rate, number of output classes), and historical context (e.g., hand-crafted features before deep learning).

Each question includes four options with one clearly correct answer and a written explanation to serve as a teaching tool rather than just an evaluation instrument. The MCQs can be used for self-assessment, peer quizzes, or as revision flashcards.

4.4. Website

As an additional deliverable, the team developed a dedicated project website to host and present all project components in a single, accessible location. The website was built as a single HTML file requiring no server or framework, making it easy to host on any static hosting platform.

Structure & Content

The site is organized into seven sections accessible via a fixed navigation bar:

- **Architecture** — an interactive layer-by-layer diagram of all 8 AlexNet layers, with hover tooltips explaining each layer's dimensions and role.
- **Key Innovations** — six cards covering ReLU, Dropout, Dual-GPU training, LRN, Data Augmentation, and Overlapping Max Pooling.
- **Presentation** — a content summary of all 24 slides grouped by topic, with a download link.
- **Demo** — the project video embedded directly from YouTube.
- **MCQs** — all 20 multiple-choice questions in an interactive accordion format, with immediate answer feedback and a live score tracker (out of 20) with a colour-coded progress bar.
- **Report** — a summary of the project methodology, challenges, and conclusions.
- **Team** — the three team members and supervisor information.

Technical Implementation

The website was implemented in a single HTML file using vanilla HTML, CSS, and JavaScript with no external dependencies beyond Google Fonts. Key technical features include: a bilingual Arabic/English toggle that switches the full site between RTL and LTR layouts; scroll-triggered reveal animations; a reading progress bar; and interactive MCQ cards that track the user's score in real time.

The website link:

5. AI Tools Used & How They Were Used

The project used AI tools as assistants at every stage. The table below documents each tool, the specific tasks it supported, and the validation steps taken to ensure the team retained full understanding and ownership of all outputs.

Tool	Used For	Validation / Human Oversight
NotebookLM (Google)	Generating the 7–8 minute AI podcast/explainer video. The team uploaded source materials and wrote a detailed prompt specifying the narrative structure, analogies, and key concepts to cover.	The generated script was reviewed line-by-line against the original AlexNet paper and course materials. Factual inaccuracies (e.g., imprecise pooling description) were corrected before finalising.
Claude (Anthropic)	Generating the 20 MCQs with explanations, and producing this report document in Word and PDF format.	Every question and explanation was checked for technical correctness. Questions with ambiguous wording were rewritten. The report structure and factual claims were verified by the team before submission.
General LLM Research	Clarifying concepts during the research phase (e.g., understanding the mathematics of LRN, the mechanics of overlapping pooling).	Used only as a starting point. All explanations were verified against the original paper and trusted sources before being included in deliverables.
Claude (Anthropic)	was used to assist in structuring and building the website	All content questions, explanations, architecture details, and report summaries were reviewed and validated by the team before inclusion. The website integrates all four project deliverables into one unified experience, fulfilling the project's optional requirement of publishing deliverables on a dedicated website.

6. Challenges Faced & How They Were Managed

6.1. Balancing depth with accessibility

Challenge: AlexNet involves several overlapping technical concepts (pooling, normalisation, regularisation, GPU parallelism) that are individually complex. Presenting them in a way that is accurate but also easy to follow in a few minutes was difficult.

How it was managed: The team adopted a 'one innovation per slide' rule in the presentation and used concrete analogies in the video (e.g., comparing dropout to students studying independently to avoid over-reliance on one person). The MCQ explanations also served as internal benchmarks: if an explanation was too technical to write clearly in 2–3 sentences, the concept needed to be simplified in the main materials too.

6.2. Validating AI-generated content

Challenge: NotebookLM and other LLMs occasionally produced plausible-sounding but slightly inaccurate statements (e.g., conflating the roles of LRN and Batch Normalisation, or overstating the memory savings from the dual-GPU setup).

How it was managed: The team read the original Krizhevsky, Sutskever, and Hinton (2012) paper directly and cross-referenced every factual claim. A checklist of 15 key facts was prepared before the video review to catch inaccuracies systematically rather than relying on casual reading.

6.3. Prompt engineering for the video

Challenge: Initial prompts to NotebookLM produced a generic overview that listed facts without connecting them. The resulting script felt like a Wikipedia summary rather than an explanation.

How it was managed: The prompt was refined to explicitly request: (a) the 'why' behind each design choice, (b) the use of one concrete analogy per concept, (c) a narrative arc from 'problem' to 'solution' to 'impact', and (d) transitions that link each concept to the next. This produced a significantly more engaging and educational output.

6.4. Deciding what to exclude

Challenge: AlexNet has many technical details (specific stride values, the exact LRN formula, padding configurations) that are important in research but would overwhelm a beginner-level audience.

How it was managed: The team defined a clear target audience (someone who understands basic neural networks but has not studied CNNs) and used this as a filter. Any detail that required more than one additional concept to explain was either moved to a 'details' slide or omitted, with a note directing interested viewers to the original paper.

7. Conclusions

This project demonstrated that a deep understanding of a technical concept can be effectively communicated through a combination of visual, interactive, and written formats. AlexNet was an ideal subject: it is historically significant, technically rich, and conceptually approachable once its innovations are explained in context rather than in isolation.

The key takeaways from the project are as follows:

- AlexNet's impact was not from any single innovation but from the combination of ReLU, Dropout, Data Augmentation, Overlapping Pooling, LRN, and GPU-accelerated training working together at a scale that had not been attempted before.
- AI tools such as NotebookLM and Claude significantly accelerated the production of deliverables, but required careful validation and thoughtful prompting to produce accurate, high-quality outputs. The quality of the prompt directly determined the quality of the output.
- The process of designing MCQ explanations was unexpectedly valuable as a self-assessment tool: it forced the team to articulate why each answer was correct, which surfaced gaps in understanding that were then addressed.
- Communicating complex ideas to a non-expert audience is a distinct skill from understanding those ideas technically. The project required the team to develop both simultaneously.
- AlexNet's limitations computational cost, large kernel sizes, Western-centric dataset bias, and eventual replacement by more efficient architectures like VGGNet and ResNet are as instructive as its successes: they show how each generation of deep learning models is designed in response to the known weaknesses of the previous one.

In conclusion, AlexNet was the spark that ignited the modern deep learning era. Understanding it deeply is not just an academic exercise it is the foundation for understanding every major CNN architecture that followed.